

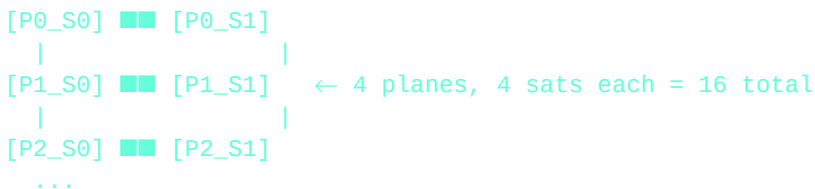
# AI-Native LEO Satellite Network — How It Works

## ■ ■ What Is This?

This is a **LEO (Low Earth Orbit) satellite constellation simulator** with AI-powered routing. It models how data packets travel across a network of **16 satellites** orbiting Earth, and uses **reinforcement learning** to find the best path depending on the mission goal.

Think of it like a **self-driving road network in space**.

## 1. The Network — What It Looks Like



- **4 Orbital Planes** (P0–P3) — like lanes in space
- **4 Satellites per Plane** (S0–S3) — evenly spaced in each orbit
- **ISL = Inter-Satellite Links** — radio links connecting neighboring satellites
- *Intra-plane links* (along the orbit): 1,500 km
- *Inter-plane links* (between orbits): 2,500 km
- Satellites **orbit at ~550 km altitude** and keep moving — the network is dynamic

## 2. The AI Brain — How Each Satellite Thinks

Every satellite is an **autonomous AI agent** running a 5-step cognitive loop:

| Step                    | What It Does                                                                          |
|-------------------------|---------------------------------------------------------------------------------------|
| ■ ■ <b>**PERCEIVE**</b> | Reads current link conditions: latency, bandwidth, packet loss, congestion            |
| ■ <b>**STATE**</b>      | Updates its internal map — notes which links just failed or recovered                 |
| ■ <b>**DECIDE**</b>     | Looks up its <b>**Q-table**</b> to find which neighbor leads to the fastest/best path |
| ■ <b>**ACT**</b>        | Forwards the packet to the chosen next satellite                                      |
| ■ <b>**LEARN**</b>      | Updates the Q-table using the Bellman equation based on what actually happened        |

## The Bellman Equation (how it learns)

$$Q(\text{hop}) \leftarrow Q(\text{hop}) + \alpha \times [\text{cost} + \gamma \times \min Q(\text{next})] - Q(\text{hop})$$

- $\alpha$  (**alpha = 0.1**) — learning rate: how fast it updates
- $\gamma$  (**gamma = 0.9**) — discount factor: how much it values future rewards
- **cost** — actual latency / packet loss experienced on that link

Over time, satellites build a **Q-table**: a lookup table that maps (**destination, neighbor**) → **expected cost**. The satellite always picks the neighbor with the **lowest Q-value** (lowest expected cost).

## 3. Mission Intent — Routing Changes Based on Goal

The same network can route differently depending on **why** the packet is being sent:

| ■ CRITICAL DISASTER RESPONSE     | ■ EARTH OBSERVATION               | ■ SECURE MISSION                  |
|----------------------------------|-----------------------------------|-----------------------------------|
| **Minimize latency**             | **Maximize throughput**           | **Maximize resilience**           |
| Fastest path, even if fewer hops | Most data through, even if slower | Avoid risky links, even if longer |
| Deadline: 50ms                   | Deadline: 500ms                   | Deadline: 200ms                   |
| Used for: emergency comms        | Used for: image downloads         | Used for: military/encrypted data |

The cost function that feeds into the Q-table changes per intent — so the **same 16 satellites route packets differently** depending on the mission tag attached to each packet.

## 4. What Happens When a Link Fails?

1. A satellite detects its ISL neighbor is unreachable (**ISLState = FAILED**)
2. It adds the failed neighbor to its **recently\_failed\_links set**
3. On the next packet forward decision, it **automatically skips** failed neighbors
4. It reroutes through alternative paths — no central control needed
5. When the link recovers, it's removed from the failed set and Q-learning resumes

*This is called **autonomous swarm recovery** — the constellation heals itself.*

## 5. Real Results (from your simulation data)

### Algorithm Comparison (100 packets, normal conditions)

| Algorithm                              | PDR      | Avg Latency | Packet Loss |
|----------------------------------------|----------|-------------|-------------|
| Dijkstra (static, pre-computed)        | **100%** | **15ms**    | 0%          |
| Basic Q-Routing                        | 58%      | 43ms        | 42%         |
| **AI-Native Q-Routing (this project)** | **94%**  | 55ms        | 6%          |

Under Link Failure (real-world stress test)

| Algorithm                    | PDR Under Failure                        | QoS Satisfied |
|------------------------------|------------------------------------------|---------------|
| Dijkstra                     | 100% (but can't reroute if route breaks) | 100%          |
| Basic Q-Routing              | 58%                                      | 48%           |
| **AI-Native (this project)** | **82%**                                  | **56%**       |

*Key insight: Dijkstra gives perfect results only if the network never changes. In space, links fail constantly. The AI-native router adapts — that's why it matters.*

Congestion Scaling (more packets = more stress)

| Traffic Load | Packets Delivered | Avg Latency |
|--------------|-------------------|-------------|
| 10 packets   | 90% (9/10)        | 40ms        |
| 50 packets   | 84% (42/50)       | 58ms        |
| 100 packets  | 89% (89/100)      | 60ms        |
| 200 packets  | 88% (176/200)     | 58ms        |

The AI router **holds steady around 85–90% PDR** even as traffic quadruples — showing it scales.

6. The 3D Visualizer — What Each Color Means

| Color              | Meaning                                       |
|--------------------|-----------------------------------------------|
| ■ Blue satellite   | Normal, idle satellite                        |
| ■ Green satellite  | **Source** — packet starts here               |
| ■ Yellow satellite | **Destination** — packet ends here            |
| ■ Red satellite    | **Failed / offline** satellite                |
| Glowing cyan line  | **Active route** (laser beam)                 |
| Moving orange dots | **Packets** traveling the route (comet trail) |
| Dim blue grid      | All ISL links in the constellation            |

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## 7. One-Line Summary to Tell Someone

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*"It's a digital twin of a satellite internet network (like Starlink) where each satellite uses AI to learn the best routes in real-time — and automatically reroutes around failures without needing a ground controller."*